Vocabulary-NB4-TOEIC Architecture

상세 아키텍처 도면. 5개 컴포넌트 (app / frontend / backend / database / ops) + 외부 시스템 (OAuth, TTS, CDN, Monitoring).

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1. 전체 시스템 개요 (Component Topology)

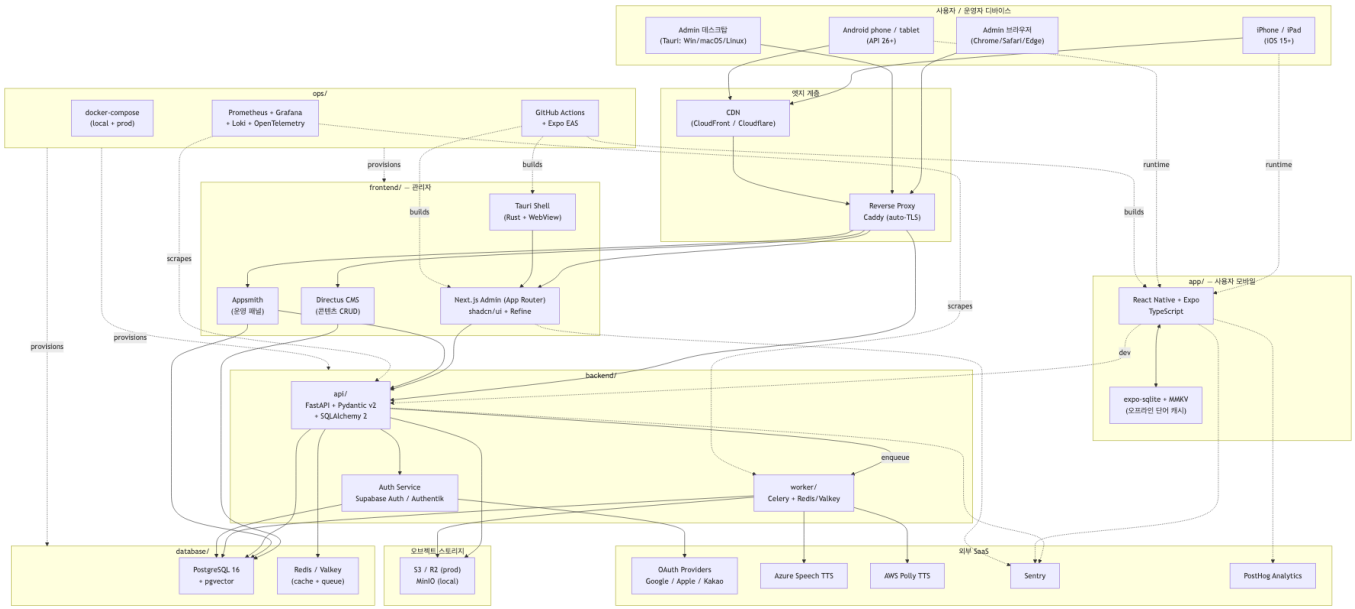


Figure 1: diagram 1

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2. 인증 플로우 (OAuth + Email OTP)

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3. 학습 데이터 흐름 (단어 카드 학습 + 진도 저장)

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4. TTS 배치 파이프라인 (예문 다국적 음원 생성)

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5. 환경 분리 (local vs prod) + 배포 토폴로지

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6. 컴포넌트 책임 매트릭스 (RACI 요약)

책임	app	frontend	backend/api	backend/worker	database	ops
사용자 인증 토큰 보관	**R**	—	A	—	C	C

책임	app	frontend	backend/api	backend/worker	database	ops
학습 카드 렌더링/재생	**R**	—	C	—	C	—
콘텐츠 CRUD	—	**R**	A	—	C	—
TTS 음원 생성	—	T (트리거)	A	**R**	C	I
진도/SRS 계산	C	C	**R**	C	C	—
스키마 마이그레이션	—	—	C	—	C	**R**
빌드/배포	—	C	C	C	C	**R**
모니터링/알림	I	I	I	I	I	**R**

R = Responsible, A = Accountable, C = Consulted, I = Informed, T = Trigger.

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7. 데이터 모델 핵심 엔터티 (논리)

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8. 보안 경계 (Trust Boundaries)

보안 원칙: - JWT 검증 후에만 데이터존 접근. - Object Storage 직접 URL 노출 금지 — presigned URL 5분 만료. - Postgres RLS로 `user_id = auth.uid()` 강제. - Secrets: ops/env (gitignore) + GitHub Actions Secret + AWS Secrets Manager (prod). - 비밀번호: Argon2id, OTP: TOTP/HMAC + 1회용.

끝.

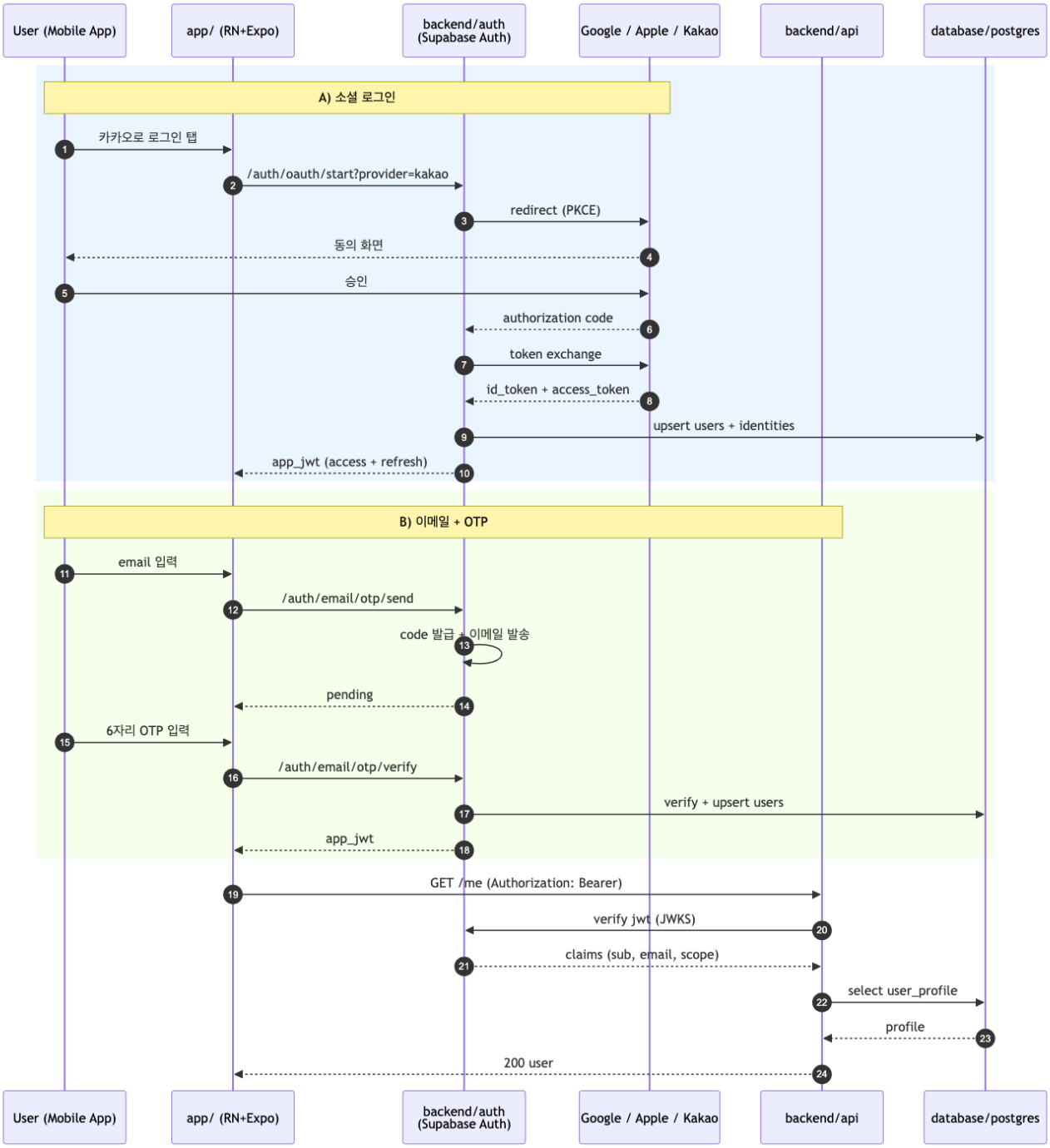


Figure 2: diagram 2

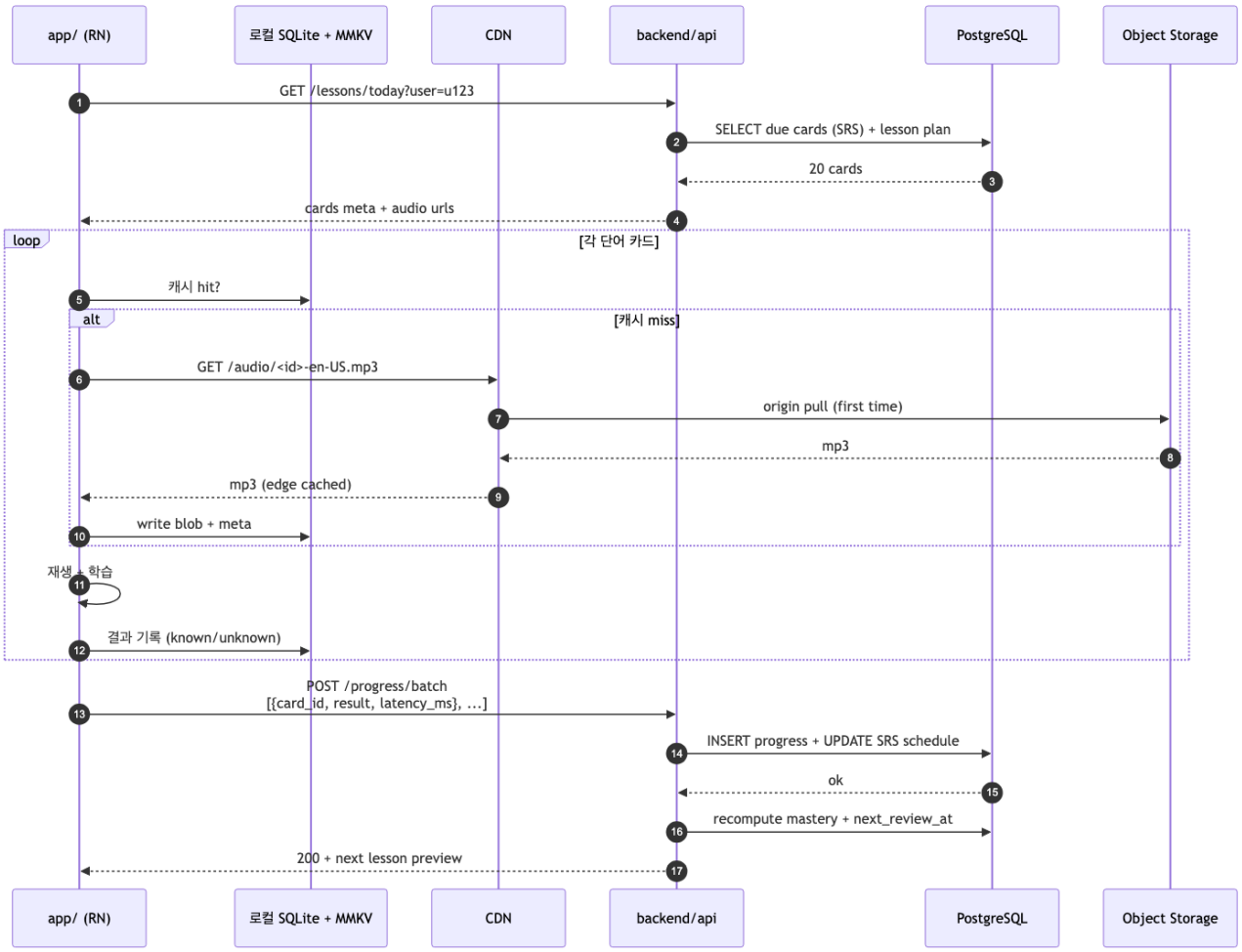


Figure 3: diagram 3

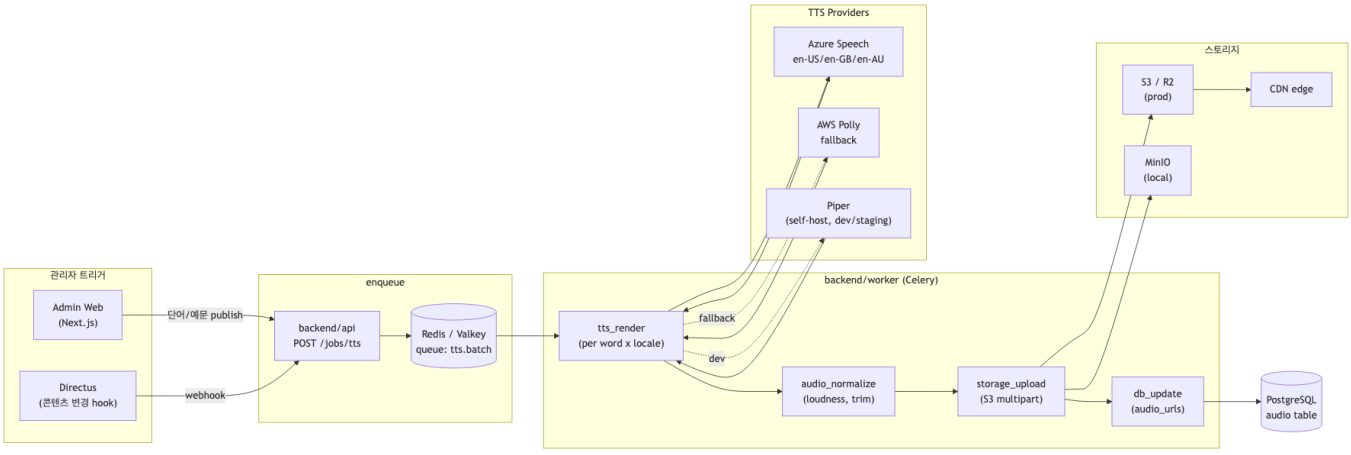


Figure 4: diagram 4

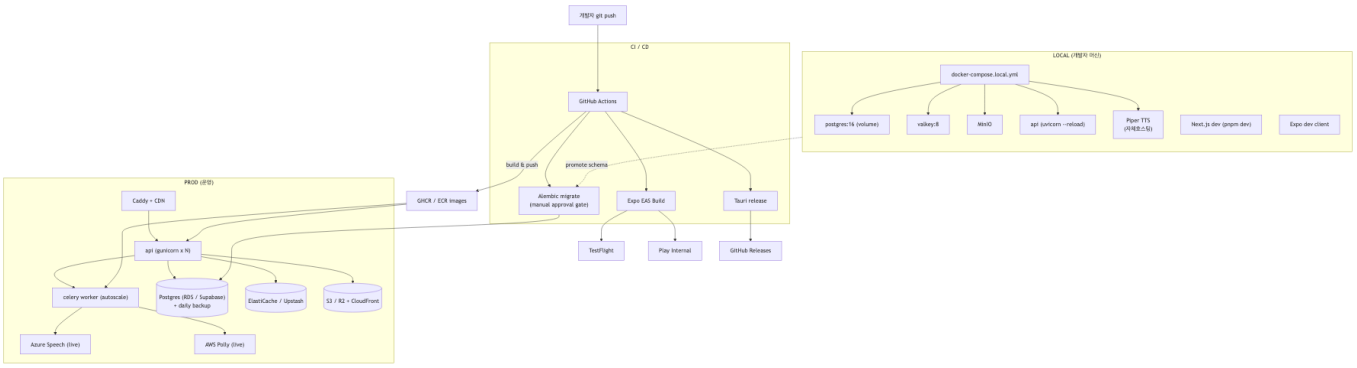


Figure 5: diagram 5

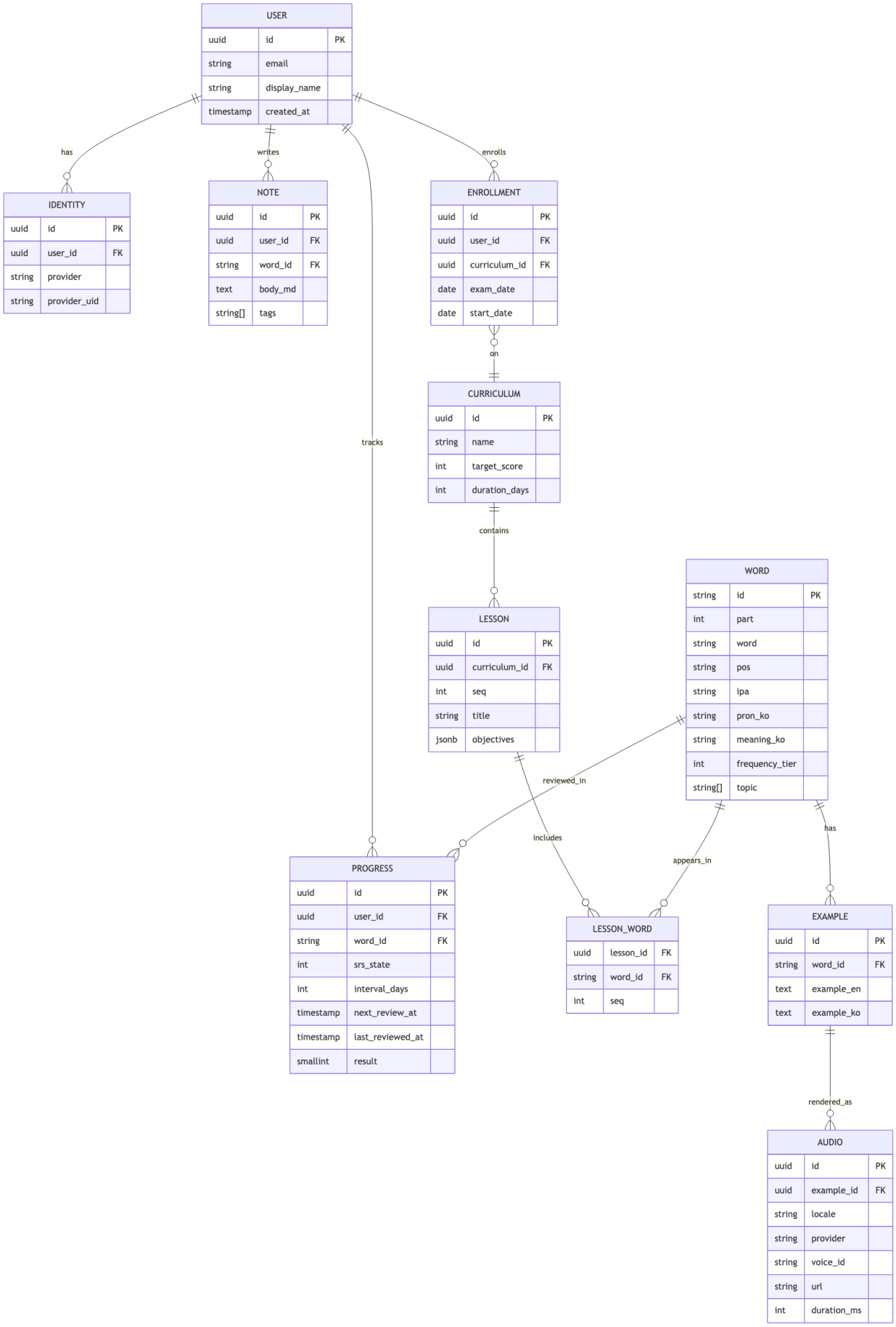


Figure 6: diagram 6

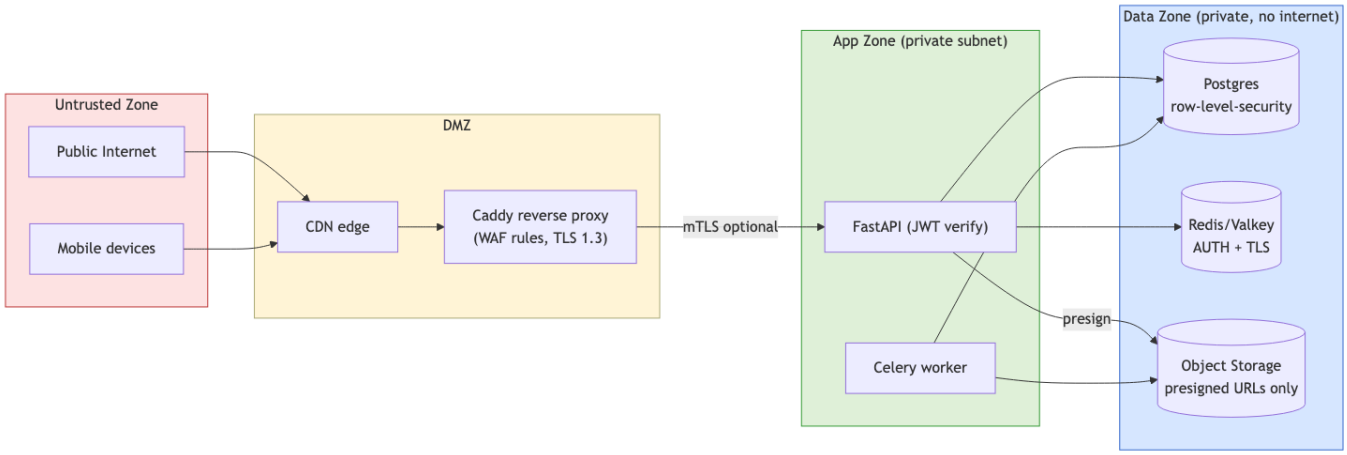


Figure 7: diagram 7